

Exact qutrit APPT purity and the unrestricted higher-dimensional asymptotic law

Yongxian Zhang*

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Abstract

We give two results on the purity of absolutely positive partial transpose (APPT) states. First, for every integer $n \geq 3$, the exact maximum on $\mathbb{C}^3 \otimes \mathbb{C}^n$ is $(3n+8)/(3n+2)^2$ for $3 \leq n \leq 8$ and $3/(8n)$ for $n \geq 9$. This proves the qutrit specialization of the maximal-value and attainment conjecture of Ahiable, Kothakonda and Winter. The entire qutrit theorem, including the physical global-unitary quantifier and actual attainment, is formalized in Lean 4.34.1 and frozen in an immutable proof release. Second, the proposed general-dimensional formula is false: an explicit APPT state on $\mathbb{C}^{10} \otimes \mathbb{C}^{38}$ has purity strictly larger than both conjectured candidates, and an infinite family gives further violations. We then prove a single corrected leading law whenever $D = mn \rightarrow \infty$, uniformly over all $2 \leq m \leq n$:

$$P_{\max}(m, n) - \frac{1}{D} \sim \frac{\max\{8, 4 + D/(m^2 - 1)\}}{D^2}.$$

For $m \rightarrow \infty$ this becomes $D^2(P_{\max} - 1/D) \sim \max\{8, 4 + n/m\}$, while fixed m gives $D(P_{\max} - 1/D) \rightarrow 1/(m^2 - 1)$. The higher-dimensional arguments are written analytic proofs supported by exact rational checkers; they are not included in the qutrit Lean certificate. Exact finite-dimensional maximizers, their classification, and the absolute-separability maximum in general dimensions remain open.

Keywords. Absolute PPT; purity; spectra; quantum states; exact certificates; Lean formalization.

1 Introduction and main result

Fix a tensor product $\mathbb{C}^3 \otimes \mathbb{C}^n$ and let $\Gamma = T \otimes \text{id}$ denote partial transposition on its first factor. A density matrix ρ is *absolutely PPT* if

$$(U\rho U^*)^\Gamma \succeq 0 \quad \text{for every } U \in \mathcal{U}(3n). \quad (1)$$

Unlike the PPT property in a fixed basis, this condition depends only on the eigenvalues. Purity, $\text{Tr}(\rho^2)$, measures the squared Hilbert–Schmidt norm. Determining its maximum is equivalent to determining the smallest ball centered at the maximally mixed state that contains the APPT set.

Hildebrand characterized the APPT property by spectral linear matrix inequalities [1]. Dung and Khoi investigated maximum purity and proposed numerical formulas for qubit–qudit and qutrit–qudit systems [2]. Their qutrit formula is not the formula proved here. Tran’s later outer-polytope

*School of Computer Science and Engineering, South China University of Technology, Guangzhou, China. Correspondence: mxymmxym1@gmail.com. ORCID: <https://orcid.org/0009-0000-3864-3536>.

analysis gives a strict qutrit upper bound that disproves that earlier qutrit conjecture [4]; we make no claim to the first disproof. Ahable, Kothakonda and Winter computed the exact maximum over an inscribed APPT polytope and conjectured that it is also the maximum over the full APPT spectral set [3, Theorem 6.3, Corollary 6.4 and Conjecture 6.7]. Its specialization to local dimension three is the following theorem.

Theorem 1.1 (Exact qutrit–qudit APPT purity). *For every integer $n \geq 3$,*

$$\max_{\rho \in \text{APPT}_{3,n}} \text{Tr}(\rho^2) = p_n := \begin{cases} \frac{3n+8}{(3n+2)^2}, & 3 \leq n \leq 8, \\ \frac{3}{8n}, & n \geq 9. \end{cases} \quad (2)$$

The first value is attained by $(I + 2vv^)/(3n+2)$ for any unit vector v . The second value is attained by $(I + P)/(4n)$ for any rank- n orthogonal projection P .*

Here $\text{APPT}_{3,n}$ denotes the trace-one positive-semidefinite matrices satisfying (1). The result includes both an upper bound for every such matrix and an actual attaining state. It confirms the value and provides attaining spectra from the conjecture of [3]; it does not classify all states attaining equality. The two displayed witnesses have spectra

$$\frac{1}{3n+2}(3, 1, \dots, 1), \quad \frac{1}{4n}(\underbrace{2, \dots, 2}_n, \underbrace{1, \dots, 1}_{2n}), \quad (3)$$

respectively. In particular, the asserted extremal values were predicted by previous work; the contribution here is their proof for the entire qutrit–qudit family and a complete physical-state formalization.

The same conjecture was stated for arbitrary local dimensions. The higher-dimensional result below shows that its proposed finite-dimensional formula cannot hold beyond the qutrit regime. We write $P_{\max}(m, n)$ for the maximum purity over all APPT density matrices on $\mathbb{C}^m \otimes \mathbb{C}^n$, where $2 \leq m \leq n$. For the conjecture, put $D = mn$ and $t = \lceil (m-1)n/2 \rceil$ and define

$$M_{m,n} = \max \left\{ \frac{D+8}{(D+2)^2}, \frac{D(m-1)^2 + 4mt}{[D(m-1) + 2t]^2} \right\}. \quad (4)$$

The second part of this paper proves that $P_{\max}(m, n) > M_{m,n}$ for explicit dimensions and gives the asymptotic replacement for $M_{m,n}$ when m grows.

The proof is computer-assisted in a precise sense. Two polynomial nonnegativity lemmas are established by rational certificates supplied as finite data. One covers the six dimensions 9, 12, $\dots$, 24; the other covers arbitrary real endpoint populations with sum at least 18. All remaining steps are analytic, linear-algebraic, or elementary algebraic. Sections 3 and 4 state the exact identities and explain how to reconstruct and check every term. The Lean proof independently verifies these identities and the bridges to the physical theorem. Neither a numerical optimization result nor a solver success flag is a premise.

2 Physical necessity and the nine selected eigenvalues

Write $D = 3n$ and order the eigenvalues of a density matrix as

$$\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_D \geq 0, \quad \sum_{i=1}^D \lambda_i = 1. \quad (5)$$

For $n \geq 3$, select the nine distinct positions

$$y = (\lambda_1, \lambda_2, \lambda_3, \lambda_{D-5}, \lambda_{D-4}, \lambda_{D-3}, \lambda_{D-2}, \lambda_{D-1}, \lambda_D). \quad (6)$$

Define the real symmetric matrices

$$A(y) = \begin{pmatrix} 2y_9 & y_8 - y_1 & y_6 - y_2 \\ y_8 - y_1 & 2y_7 & y_5 - y_3 \\ y_6 - y_2 & y_5 - y_3 & 2y_4 \end{pmatrix}, \quad (7)$$

$$B(y) = \begin{pmatrix} 2y_9 & y_8 - y_1 & y_7 - y_2 \\ y_8 - y_1 & 2y_6 & y_5 - y_3 \\ y_7 - y_2 & y_5 - y_3 & 2y_4 \end{pmatrix}. \quad (8)$$

These are the familiar qutrit spectral tests, with the two matrices possibly interchanged relative to another labeling convention.

Lemma 2.1 (Physical necessity). *If ρ is APPT, then $A(y) \succeq 0$ and $B(y) \succeq 0$ for y in (6). More generally, the conclusion holds for any nine distinct eigenvalue positions assigned to the coordinates of y .*

Proof. By a global unitary, first diagonalize ρ and move the selected eigenvalues into a 3×3 product-basis corner. For A , assign those eigenvalues to the corner coordinates according to the array

$$\begin{pmatrix} 9 & 8 & 6 \\ 1 & 7 & 5 \\ 2 & 3 & 4 \end{pmatrix}; \quad \text{for } B \text{ use } \begin{pmatrix} 9 & 8 & 7 \\ 1 & 6 & 5 \\ 2 & 3 & 4 \end{pmatrix}.$$

An entry j in an array means that the corresponding diagonal entry is y_j . The other eigenvalues occupy the complementary coordinates. On each of the three planes spanned by $e_i \otimes e_j, e_j \otimes e_i$ with $i < j$, apply the real unitary

$$\frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix},$$

and leave the diagonal corner coordinates and the orthogonal complement fixed. A diagonal pair with upper-coordinate value s and lower-coordinate value a acquires off-diagonal entries $(s - a)/2$. Partial transposition carries these entries to the matrix elements between $e_i \otimes e_i$ and $e_j \otimes e_j$. Thus the principal submatrix on the three diagonal corner vectors is exactly $A(y)/2$ or $B(y)/2$. Condition (1) makes the whole partial transpose positive semidefinite; its principal submatrices are positive semidefinite as well. $\square$

This argument only uses necessity. It does not assume that these matrices characterize APPT, and it does not need the converse direction of Hildebrand's criterion. In Lean, arbitrary injections of nine eigenvalue slots are extended to permutations of the full index type, the corner unitary is extended by the identity, and its entries are verified explicitly. Hermitian diagonalization then supplies the actual ordered spectrum, together with

$$\text{Tr}(\rho) = \sum_i \lambda_i, \quad \text{Tr}(\rho^2) = \sum_i \lambda_i^2. \quad (9)$$

These equations are proved from diagonalization, not supplied as spectral-interface hypotheses.

We shall use the following immediate PSD consequences. For $C = A(y)$ or $B(y)$, its determinant, every principal 2×2 minor and every real quadratic form $v^T C v$ are nonnegative. Also,

Lemma 2.2. *If $y_1 \geq \dots \geq y_9 \geq 0$ and $A(y) \succeq 0$, then $2y_4 - y_3 \geq 0$.*

Proof. Writing $\mathbf{1} = (1, 1, 1)^T$, we have

$$\frac{1}{2}\mathbf{1}^T A(y)\mathbf{1} = \sum_{j=4}^9 y_j - \sum_{j=1}^3 y_j \geq 0.$$

Ordering gives $3y_3 \leq \sum_{j=1}^3 y_j \leq \sum_{j=4}^9 y_j \leq 6y_4$. □

3 Six finite cubic certificates

For $D \in \{9, 12, 15, 18, 21, 24\}$ introduce gaps

$$g_i = \lambda_i - \lambda_{i+1} \ (i < D), \quad g_D = \lambda_D.$$

Then $g_i \geq 0$ and $\lambda_i = \sum_{j=i}^D g_j$. Conversely every nonnegative gap vector defines an ordered nonnegative spectrum in this way. Set

$$T = \sum_{i=1}^D \lambda_i, \quad S = \sum_{i=1}^D \lambda_i^2,$$

and form y using (6).

Lemma 3.1 (Finite certificate identities). *For each of the six dimensions above, the ancillary certificate specifies positive rational numbers $c_{D,\alpha}$ and explicitly named homogeneous cubic polynomials $F_{D,\alpha}$ such that the polynomial identity*

$$((D+8)T^2 - (D+2)^2 S)T = \sum_{\alpha} c_{D,\alpha} F_{D,\alpha}(g) \tag{10}$$

holds over $\mathbb{Q}[g_1, \dots, g_D]$. Every polynomial on the right is nonnegative whenever $g \geq 0$, $A(y) \succeq 0$ and $B(y) \succeq 0$.

Certificate proof. The allowable terms are products of nonnegative gaps with PSD determinants, principal minors or quadratic forms, and plain gap monomials. Explicitly, the data use terms of the forms

$$\det C, \quad (C_{ii}C_{jj} - C_{ij}C_{ji})g_k, \quad (v^T C v)g_k g_\ell, \quad g_i g_j g_k, \quad C \in \{A(y), B(y)\}, \tag{11}$$

with the rational vector v specified in the term name. Each term is nonnegative under the hypotheses. The six rational coefficient lists are included as exact decimal-free strings in the ancillary JSON files. Table 1 specifies their sizes and the resulting bounds.

For verification, substitute the linear tail sums for λ , expand both sides of (10) over the rational numbers, and compare the coefficient of every monomial. The supplied checker performs precisely this operation using Python integers and `Fraction`; it also verifies strict positivity of every certificate weight. All six coefficient differences vanish. Equivalently, one may clear denominators by the least common multiple of the denominators and compare integer coefficient dictionaries. The Lean finite modules separately prove the same identities through small symbolic expansions and kernel-checked integer coefficient merges, and derive their nonnegativity from the PSD consequences described above. The data, not a numerical search procedure, constitute the finite part of the proof. □

Corollary 3.2. *Every APPT density matrix on $\mathbb{C}^3 \otimes \mathbb{C}^n$ with $3 \leq n \leq 8$ satisfies $\text{Tr}(\rho^2) \leq (3n+8)/(3n+2)^2$.*

Proof. Lemma 2.1 supplies the two PSD hypotheses. Ordering supplies nonnegative gaps, and trace one gives $T = 1$. Hence (10) yields $(D+8) - (D+2)^2 S \geq 0$. Apply (9). □

n	D	Certificate terms	$(D+8)/(D+2)^2$	Data filename suffix
3	9	166	17/121	cubic_certificate_search.json
4	12	363	5/49	D12_cubic_certificate.json
5	15	679	23/289	D15_fast_certificate.json
6	18	1139	13/200	D18_fast_certificate.json
7	21	1769	29/529	D21_fast_certificate.json
8	24	2596	8/169	D24_fast_certificate.json

Table 1: Finite exact certificates. Each filename has the prefix `qutrit_`. Counts refer to positive weighted terms, not expanded monomials.

4 A uniform bound by endpoint compression

The number of middle eigenvalues grows with n , but their positions do not enter the two PSD matrices. The following elementary fact removes their individual values.

Lemma 4.1 (Endpoint compression). *Let $b \leq x_i \leq a$ for $1 \leq i \leq M$, where $b \leq a$. There exist real $t, z \geq 0$ such that*

$$t + z = M, \quad \sum_i x_i = ta + zb, \quad \sum_i x_i^2 \leq ta^2 + zb^2. \quad (12)$$

Proof. For $a > b$, choose

$$t = \frac{\sum_i x_i - Mb}{a - b}, \quad z = M - t.$$

The interval constraints imply $0 \leq t \leq M$. Since $(a - x_i)(x_i - b) \geq 0$, we have $x_i^2 \leq (a + b)x_i - ab$; sum this inequality to obtain (12). If $a = b$, take $t = M$ and $z = 0$. $\square$

The populations t, z are real rather than necessarily integral. This is important: rounding them would not preserve the trace exactly.

For an arbitrary ordered $y \in \mathbb{R}_{\geq 0}^9$ and $t, z \geq 0$, write

$$T(y, t, z) = \sum_{i=1}^9 y_i + ty_3 + zy_4, \quad (13)$$

$$S(y, t, z) = \sum_{i=1}^9 y_i^2 + ty_3^2 + zy_4^2. \quad (14)$$

Take $g_i = y_i - y_{i+1}$ for $i < 9$ and $g_9 = y_9$, and abbreviate

$$w = t + z - 18, \quad h = ty_3 - zy_4, \quad q = 9T - 4(9 + t + z)(y_3 + y_4). \quad (15)$$

Lemma 4.2 (Uniform rational certificate). *For every ordered nonnegative y , every $t, z \geq 0$ with $t + z \geq 18$, and $A(y), B(y) \geq 0$,*

$$(9T(y, t, z)^2 - 8(9 + t + z)S(y, t, z))T(y, t, z) \geq 0. \quad (16)$$

In particular, if $T(y, t, z) = 1$, then

$$S(y, t, z) \leq \frac{9}{8(9 + t + z)}. \quad (17)$$

Certificate proof. The file `qutrit_uniform_parameter_search.json` gives 1,635 positive rational weights c_α and explicitly named polynomials F_α satisfying

$$(9T^2 - 8(9 + t + z)S)T = \sum_{\alpha=1}^{1635} c_\alpha F_\alpha(g, t, z). \quad (18)$$

This is a polynomial identity in the nine gaps and the two real parameters. Its nonnegative generators consist of the PSD-based and gap terms in (11), the additional terms $(2y_4 - y_3)g_i g_j$, nonnegative monomials in the gaps and t, z , and terms of the form

$$h^2 g_i \xi \quad \text{or} \quad (aq + bh)^2 g_i \xi, \quad \xi \in \{1, t, z, w\}, \quad a, b \in \mathbb{Z}. \quad (19)$$

The data also multiply PSD-based or ratio terms by explicitly specified products of t, z, w . The exact term names determine which products and integers are used; all are evaluated with $w = t + z - 18$. Every factor is nonnegative under the hypotheses: for the ratio factor use Lemma 2.2, and for the squared factors no sign assumption on h or q is necessary.

The ancillary checker reconstructs the identity exactly, including the substitutions (15), and confirms equality of every rational coefficient. The fixed Lean module `APPT.Uniform` proves (16) using the same exact data; its dependencies supply evaluation-preserving integer coefficient operations and sign proofs. With $T = 1$, divide by the positive number $8(9 + t + z)$ to obtain (17). $\square$

Proposition 4.3 (All large dimensions). *For every integer $n \geq 9$, every APPT density matrix on $\mathbb{C}^3 \otimes \mathbb{C}^n$ has purity at most $3/(8n)$.*

Proof. Let $D = 3n$, choose y as in (6), and let x contain the $M = D - 9$ middle entries $\lambda_4, \dots, \lambda_{D-6}$. Then $y_4 \leq x_i \leq y_3$ and $M \geq 18$. By Lemma 4.1, there are $t, z \geq 0$ with $t + z = M$ for which

$$T(y, t, z) = \sum_{i=1}^D \lambda_i = 1, \quad \sum_{i=1}^D \lambda_i^2 \leq S(y, t, z).$$

Lemma 2.1 supplies $A(y), B(y) \succeq 0$. The uniform certificate therefore gives

$$\text{Tr}(\rho^2) = \sum_{i=1}^D \lambda_i^2 \leq \frac{9}{8(9 + t + z)} = \frac{9}{8D} = \frac{3}{8n}.$$

The argument treats every integer $n \geq 9$ through one real-parameter identity; it is not an extrapolation from finitely many dimensions. $\square$

5 Actual APPT attaining states

We provide physical witnesses without relying on the sufficiency direction of a spectral APPT criterion.

Lemma 5.1 (A rank-one perturbation). *For every unit vector $v \in \mathbb{C}^3 \otimes \mathbb{C}^n$, the matrix $I + 2vv^*$ is APPT before trace normalization.*

Proof. A unit vector has Schmidt coefficients $s_1, s_2, s_3 \geq 0$ with $\sum_i s_i^2 = 1$, allowing zero coefficients. The nonzero eigenvalues of $(vv^*)^\Gamma$ are s_i^2 and $\pm s_i s_j$ for $i < j$. Since $2s_i s_j \leq s_i^2 + s_j^2 \leq 1$, we have $I + 2(vv^*)^\Gamma \succeq 0$. For every global unitary U , the vector Uv is again a unit vector, so the same argument applies to $U(I + 2vv^*)U^* = I + 2(Uv)(Uv)^*$. The formalization proves the same rank-one bound by an explicit skew-Gram identity and a rectangular-contraction argument instead of assuming a Schmidt-decomposition interface. $\square$

Lemma 5.2 (A positive contraction). *If $0 \preceq P \preceq I$ on $\mathbb{C}^3 \otimes \mathbb{C}^n$, then $P^\Gamma + I \succeq 0$, and $I + P$ is APPT before normalization.*

Proof. Let E_{ij} be the matrix units on $\mathbb{C}^3$, let $D_i = E_{ii}$, and for $i < j$ put $S_{ij} = E_{ij} + E_{ji}$ and $K_{ij} = E_{ji} - E_{ij}$. Tensor these matrices with I_n when they act on the full space. Direct block multiplication gives the exact identity

$$P^\Gamma + I = \sum_{i=1}^3 D_i P D_i^* + \frac{1}{2} \sum_{i < j} (S_{ij} P S_{ij}^* + K_{ij} (I - P) K_{ij}^*). \quad (20)$$

Every summand is PSD. For a global unitary U , the conjugate $Q = U P U^*$ is another positive contraction, so $I + Q^\Gamma \succeq 0$. This is exactly the APPT condition for $I + P$. $\square$

Proof of Theorem 1.1. The upper bounds are Corollary 3.2 and Proposition 4.3. For the first witness, $R = vv^*$ satisfies $R^2 = R$ and $\text{Tr } R = 1$. Hence $I + 2R \succeq 0$, $\text{Tr}(I + 2R) = D + 2$, and

$$\text{Tr} \left(\left(\frac{I + 2R}{D + 2} \right)^2 \right) = \frac{\text{Tr}(I + 8R)}{(D + 2)^2} = \frac{D + 8}{(D + 2)^2}.$$

Lemma 5.1 gives APPT. For the second witness choose a rank- n orthogonal projection P , for example the coordinate projection onto $e_1 \otimes \mathbb{C}^n$. Then $P^2 = P$, $\text{Tr } P = n$, and Lemma 5.2 gives APPT. Its trace normalization is $D + n = 4n$, and

$$\text{Tr} \left(\left(\frac{I + P}{4n} \right)^2 \right) = \frac{\text{Tr}(I + 3P)}{16n^2} = \frac{6n}{16n^2} = \frac{3}{8n}.$$

Positive scalar normalization preserves APPT. Thus each upper bound is attained by a genuine density matrix. $\square$

Corollary 5.3 (Circumradius). *The squared radius of the smallest Hilbert–Schmidt ball centered at $I/(3n)$ containing the APPT density matrices is*

$$R_n^2 = \begin{cases} \frac{4(3n-1)}{3n(3n+2)^2}, & 3 \leq n \leq 8, \\ \frac{1}{24n}, & n \geq 9. \end{cases}$$

Proof. For a Hermitian trace-one matrix, $\|\rho - I/D\|_{\text{HS}}^2 = \text{Tr}(\rho^2) - 1/D$. Substitute Theorem 1.1. This is an algebraic corollary of the formalized purity result; the release does not export a separately named Hilbert–Schmidt circumradius theorem. $\square$

6 A higher-dimensional counterexample

The general candidate (4) is the purity maximum of the inscribed spectral polytope in [3]; Conjecture 6.7 there asserts that it is also the maximum over all APPT spectra. We now give an actual APPT state outside that polytope whose purity is larger.

Theorem 6.1 (Explicit failure of the general formula). *On $\mathbb{C}^{10} \otimes \mathbb{C}^{38}$, the diagonal state*

$$\rho_{10,38} = \frac{\text{diag}(41, \underbrace{17, \dots, 17}_{146}, \underbrace{15, \dots, 15}_{233})}{6018} \quad (21)$$

is APPT and satisfies

$$\mathrm{Tr}(\rho_{10,38}^2) = \frac{2675}{1006009} > M_{10,38}.$$

More precisely, the two candidate values are $97/36481$ and $5/1881$, and the exact gaps are respectively

$$\frac{3802}{36700214329}, \quad \frac{1630}{1892302929}.$$

Consequently Conjecture 6.7 of [3] is false in its stated arbitrary-dimensional form.

Proof. We prove a slightly more general all-unitary lemma. Let $m \geq 4$, $n \geq m$, $D = mn$, let P be an orthogonal projection on $\mathbb{C}^m \otimes \mathbb{C}^n$, and let v be a unit vector in the range of P . Set

$$A = (2m - 5)I + 2P + 4(m - 4)|v\rangle\langle v|.$$

For a unit vector ψ with Schmidt coefficients $s_1 \geq \dots \geq s_m \geq 0$, put $W = (|\psi\rangle\langle\psi|)^\Gamma$. The negative eigenvalues of W are $-s_i s_j$ for $i < j$, and its least eigenvalue is $-s_1 s_2$. Since $0 \leq UPU^* \leq I$, spectral evaluation gives

$$\mathrm{Tr}(UPU^*W) \geq -\sum_{i < j} s_i s_j, \quad \langle Uv, WUv \rangle \geq -s_1 s_2.$$

Thus, for every global unitary U ,

$$\begin{aligned} \langle \psi, (UAU^*)^\Gamma \psi \rangle &\geq (2m - 5) \sum_i s_i^2 - 2 \sum_{i < j} s_i s_j - 4(m - 4)s_1 s_2 \\ &= (2m - 6)(s_1 - s_2)^2 + 2 \sum_{3 \leq i < j \leq m} (s_i - s_j)^2 + (s_1 + s_2 - \sum_{i=3}^m s_i)^2 \geq 0. \end{aligned}$$

The identity is obtained by expanding both sides, so A is APPT. If $\mathrm{rank} P = k$, the eigenvalues of A are

$$6m - 19 \text{ (1 time)}, \quad 2m - 3 \text{ (} k - 1 \text{ times)}, \quad 2m - 5 \text{ (} D - k \text{ times)}.$$

For $m = 10, n = 38, k = 147$ this is exactly (21). Its trace-square is $96300/6018^2$, which reduces to $2675/1006009$. Substitution of $m = 10, n = 38, t = 171$ into (4) gives the two displayed candidates and the stated positive rational differences. No numerical optimization or sampled unitary is used. $\square$

The counterexample is not confined to a singular boundary. If $\varepsilon = 1/1000$ and

$$\rho_\varepsilon = (1 - \varepsilon)\rho_{10,38} + \varepsilon I/380,$$

then every global-unitary partial transpose is bounded below by $I/380000$, while its purity still exceeds both candidates. The same construction gives an explicit infinite family for every $m \geq 11$ with $n = 4m$; the exact factorization and multiplicities are included in the supplementary proof file.

7 The corrected unrestricted asymptotic law

The finite counterexample leads to a sharp asymptotic replacement for the false formula. The result concerns the regime in which the smaller local dimension grows; it does not assert an exact finite-dimensional maximum.

Theorem 7.1 (Uniform APPT excess-purity asymptotic). *Let $P_{\max}(m, n)$ be the maximum purity over APPT density matrices on $\mathbb{C}^m \otimes \mathbb{C}^n$. Then, as $m \rightarrow \infty$, uniformly over integers $n \geq m$,*

$$\sup_{n \geq m} \left| \frac{D^2 (P_{\max}(m, n) - 1/D)}{\max\{8, 4 + n/m\}} - 1 \right| \rightarrow 0, \quad D = mn. \quad (22)$$

In particular, for every fixed $\gamma \geq 1$ and every sequence $n/m \rightarrow \gamma$,

$$D^2 \left(P_{\max}(m, n) - \frac{1}{D} \right) \rightarrow \max\{8, 4 + \gamma\},$$

and $P_{\max}(m, m) = m^{-2} + 8m^{-4} + o(m^{-4})$.

Proof. We give the two estimates that are uniform in the spectrum and in n . Order the eigenvalues of an APPT state as $\lambda_1 \geq \dots \geq \lambda_D$, put $R = m(m-1)/2$, $S = m(m+1)/2$, $j = D - S + 1$, $b = \lambda_j$, and $\delta_i = \lambda_i - \lambda_{D+1-i}$. For every assignment of the nonnegative δ_i to the edges of K_m , a Schmidt-vector partial-transpose test gives

$$\lambda_{\max}(A_\delta) \leq 2b, \quad (23)$$

where A_δ is the resulting weighted adjacency matrix. This uses the top R eigenvalues on antisymmetric pair vectors, the bottom R on symmetric pair vectors and the next m eigenvalues on diagonal vectors; those slots are distinct because $m^2 \leq D$.

Place the largest $m-1$ gaps on a star. Equation (23) gives $\sum_{i=1}^{m-1} \delta_i^2 \leq 4b^2$. In a second, row-by-row placement, subtract the smallest gap $w = \delta_R$ from every edge and denote the residuals by $\eta_i = \delta_i - w$. The elementary row-degree inequality

$$\sum_{i=m}^R \eta_i^2 \leq \|A_\eta m^{-1/2} \mathbf{1}\|^2$$

together with (23) yields

$$V := \sum_i \lambda_i^2 - 1/D \leq C(m, n)b^2, \quad C(m, n) = \max \left\{ 8, 4 + \frac{D}{(m-1)^2} \right\}.$$

The derivation uses the midpoint of λ_R and λ_{D-R+1} ; all remaining eigenvalues lie within $w/2$ of that midpoint. Since $jb \leq 1$ and the first j eigenvalues contribute at least $j(b - 1/D)^2$ to V whenever $b > 1/D$, this gives

$$D^2 V \leq \frac{C(m, n)}{(1 - \sqrt{C(m, n)}/j)^2}.$$

As $m \rightarrow \infty$, the denominator tends uniformly to one and $C(m, n) \leq [m/(m-1)]^2 \max\{8, 4 + n/m\}$.

For the matching lower bound, two explicit APPT constructions are used. A one-spike/plateau state gives, for every finite $m \geq 4$ and $n \geq m$, a centered coefficient tending uniformly to $4 + n/m$. A finite lacunary hierarchy of nested spectral plateaus gives a coefficient arbitrarily close to 8, with a fixed positive slack in the all-unitary graph bound before the dimension limit. The hierarchy is first held fixed, then m tends to infinity, and only afterward are the slack and number of levels optimized. Thus for every $\varepsilon > 0$, all sufficiently large m admit APPT states with

$$D^2(P_{\max} - 1/D) \geq (1 - \varepsilon) \max\{8, 4 + n/m\}$$

uniformly in $n \geq m$. Combining the upper and lower estimates proves (22). The complete graph-limit and forest argument for the lacunary construction is written in the supplementary proof files, with exact rational regression checks; no finite numerical test is used as a premise. $\square$

Remark 7.2 (Status of the higher-dimensional argument). Theorem 7.1 is a written analytic proof supported by exact rational and integer checkers. It is not part of the qutrit Lean release. It does not determine exact finite-dimensional maxima, classify maximizing spectra, or prove equality between APPT and absolutely separable states.

Theorem 7.3 (All-regime equivalent). *Let $P_{\max}(m, n)$ be as above and define*

$$A(m, n) = \frac{\max\{8, 4 + D/(m^2 - 1)\}}{D^2}, \quad D = mn.$$

Then

$$\lim_{N \rightarrow \infty} \sup_{2 \leq m \leq n, mn \geq N} \left| \frac{P_{\max}(m, n) - 1/(mn)}{A(m, n)} - 1 \right| = 0. \quad (24)$$

For each fixed $m \geq 2$, this implies

$$D P_{\max}(m, n) \rightarrow \frac{m^2}{m^2 - 1}, \quad P_{\max}(m, n) - \frac{1}{D} \sim \frac{1}{(m^2 - 1)D} \quad (n \rightarrow \infty).$$

Proof. Theorem 7.1 proves the uniform statement when the smaller dimension tends to infinity. It remains to treat a bounded subsequence of m . Fix $m \geq 2$ and use the notation $R = m(m - 1)/2$, $S = m(m + 1)/2$, $j = D - S + 1$, and $b = \lambda_j$ for an ordered APPT spectrum. The maximally entangled Schmidt-rank- m test, with the largest R eigenvalues placed on its negative witness space, gives

$$\sum_{i=1}^R \lambda_i \leq \sum_{i=D-S+1}^D \lambda_i,$$

so $\lambda_R \leq [(m+1)/(m-1)]b$. A single-edge test gives $\lambda_1 - \lambda_D \leq 2b$, hence $\lambda_1 \leq 3b$. The $N_0 = D - m^2 + 2$ eigenvalues from indices R through $D - S + 1$ therefore lie in $[b, rb]$, where $r = (m + 1)/(m - 1)$. For numbers in this interval,

$$x^2 \leq (r + 1)bx - rb^2.$$

After summing and optimizing the resulting quadratic in $z = N_0 b/T$, one obtains

$$\sum_i \lambda_i^2 \leq \frac{m^2/(m^2 - 1)}{D - m^2 + 2} + \frac{9(m^2 - 2)}{(D - S + 1)^2}.$$

This gives the fixed- m upper limit after multiplication by D . For the matching lower bound, let $c = 2/(m - 1)$, choose a rank- k projection with $k = \lceil D/(c + 2) \rceil$, and normalize $I + cP$. The sum of the negative Schmidt eigenvalues is at most $(m - 1)/2$, so this state is APPT for every global unitary. Its exact purity calculation gives

$$D \operatorname{Tr}(\rho^2) \rightarrow \frac{(c + 2)^2}{4(c + 1)} = \frac{m^2}{m^2 - 1}.$$

Thus the fixed- m limit follows. Finally, if (24) failed, a sequence with $D \rightarrow \infty$ and a fixed error would have either a bounded subsequence of m (handled by the fixed- m argument) or $m \rightarrow \infty$ (handled by Theorem 7.1). In the latter case $D/(m^2 - 1)$ and n/m differ by a factor tending to one. This sequential criterion proves (24). The detailed fixed- m derivation and exact deficit identity are recorded in the supplementary proof file. $\square$

8 Qutrit certificate data and complete formal verification

8.1 Data specification and independent checking

The ancillary directory contains all seven JSON files listed in Appendix A, copied byte for byte from the fixed formalization. Each contains a `terms` list. Its entries have a `coefficient` field holding a rational string and a `name` field specifying the polynomial. The checker reconstructs the tail sums, matrices (7)–(8), their determinants, principal minors and quadratic forms. It then expands (10) or (18) and verifies coefficient equality. The reconstruction code uses only exact integers and rational numbers; no optimizer is imported. The included input marker `exact_certificate` is checked but is not accepted as proof: coefficient equality is recomputed.

The JSON term language uses zero-based indices in explicit exponent lists and `gprod` tuples, and one-based subscripts in names such as `Aminor02*g23`; the latter means the principal minor on matrix indices 0, 2 multiplied by the twenty-third gap. The supplied reconstruction code is the unambiguous machine-readable specification of these conventions. In the uniform file, `param` entries use 0, 1, 2 to denote $t, z, t + z - 18$. All term signs are justified as in the two certificate lemmas; there is no search over or sampling of real parameter values in the verification.

From the preprint source root, the independent rational check is

```
python3 ancillary/check_certificates.py
```

The complete formalization is available separately in the immutable release [6]. Its source manifest fixes both the coefficient data and the Lean proofs. This separation keeps the preprint source archive small while preserving a precise reference to every formal dependency.

8.2 Formal statement and dependency map

The final Lean theorem in namespace `APPT.Quantum` has the form

```
theorem appt_purity_maximum_formula (n : Nat) (hn : 3 <= n) :
  IsGreatest (attainablePurities n)
    (if n <= 8 then (3*(n : Real)+8)/(3*(n : Real)+2)^2
     else 3/(8*(n : Real)))
```

Here `attainablePurities` quantifies over actual matrices indexed by `Fin 3 × Fin n`; `IsDensity` means PSD and complex trace one; `AbsolutelyPPT` quantifies over the actual complex unitary group; and `purity A` is the real part of the actual trace of A^*A . For a density matrix the spectral-moment theorem identifies this real quantity with the usual real purity.

Proof obligation	Principal Lean modules or declarations
Global-unitary corner necessity	<code>Quantum.CornerNecessity</code> , <code>Quantum.SpectralNecessity</code>
Ordered eigenvalues and moments	<code>Quantum.SpectralData</code> , <code>Quantum.SpectralMoments</code>
Six finite certificate bounds	<code>Finite9Bound</code> through <code>Finite24Bound</code> , and <code>SpectrumBound*</code>
Uniform and middle-spectrum bound	<code>Uniform</code> , <code>Compression</code> , <code>UniformBridge</code> , <code>OrderedSpectrum</code>
Physical APPT witnesses	<code>Quantum.ShortAttainment</code> , <code>Quantum.LongAttainment</code> , <code>Quantum.Attainment</code>
All- n maximum	<code>Quantum.Maximum</code> : <code>appt_purity_maximum_formula</code>

8.3 Build, replay, and rejection controls

The proof uses Lean 4.34.1 and Mathlib commit

d13f23b723b8a846827a245b89c10fc7d3f11612.

The published proof commit is 2fc3f25179cf1128c4bccbde65fd3a07c093ad6d. The final successful verification run is 37968137659, at verified source commit

f372047c63ba501b13301b4b3ba089c261e3722e.

All 942 pinned proof, tool and data files agree between the verified and published sources. The mathematical theorem files are unchanged from the first successful full proof replay; later corrections concerned auxiliary tests and verification orchestration.

All 879 local modules and the default Lake target passed. The final audit replayed the transitive closure of 58 roots, including the exact maximum theorem: 155,787 declarations were checked in an initially empty Lean kernel at trust level zero. The only axioms in the audited closure are `propext`, `Classical.choice`, and `Quot.sound`. The checker rejects unsafe or partial mathematical dependencies and unapproved axioms, and compares each replayed root’s type and universe parameters with the original. No `sorry`, `admit`, custom axiom or untrusted `native_decide` is used in the mathematical closure.

The paired failure controls are part of the record. They alter a coefficient, a PSD-corner example, or an endpoint value, and require the expected rejection rather than an arbitrary process failure. A further test replaces only the proof of the same final maximum theorem by `True.intro`, preserving its name, type and universe parameters. The changed declaration’s 8,581 actual dependencies are positively checked first; a fresh trust-zero kernel then rejects the forged theorem for a declaration type mismatch. The large original proof has already passed the full positive replay. Dependencies belonging only to the discarded original proof are not unnecessarily repeated in the negative test.

The final workflow reused independently compiled components bound to exact source, object and log hashes. It does *not* claim a fresh build of all 879 modules on a single runner. This does not change the independent empty-kernel replay of the full mathematical closure. Each recorded local-module build had a 300-second limit; the largest recorded time was 239.703 seconds. Small coefficient lemmas, literal intermediate coefficient lists and explicit integer constructors avoid an unbounded monolithic elaboration.

The fresh reproduction command, from the formalization’s package root, is

```
lake exe cache get
python3 scripts/source_manifest.py --check
python3 reproduce.py --fresh --jobs 2
```

The fresh option removes the project’s own build directory, not the external Mathlib caches. Literal outputs, source hashes, component provenance, dependency inventories and the earlier failed audit attempts are retained. A failed historical workflow is never relabeled as a successful one.

9 An analytic corollary for absolute separability

A state is *absolutely separable* if every global-unitary conjugate is separable across the fixed bipartition. Since separable states are PPT, the absolutely separable set is contained in the APPT set. The following consequence concerns its extremal purity, not equality of the two sets.

Corollary 9.1 (Absolute-separability purity). *For every $n \geq 3$, the maximum purity of absolutely separable states on $\mathbb{C}^3 \otimes \mathbb{C}^n$ equals p_n in (2).*

Proof. The APPT theorem supplies the upper bound. For the large branch, use the sufficient condition of Kondra, Barrios Hita, Neumann, Kampermann and Bruß [5, Supplemental Material, Lemma 2]: in local dimensions $m \leq n$, a full-rank state with

$$\lambda_{\max}/\lambda_{\min} \leq (m+1)/(m-1)$$

is separable. As the criterion is spectral, it also certifies absolute separability. The state $(I+P)/(4n)$ has eigenvalue ratio 2, exactly this threshold for $m=3$.

For the small branch, we give a direct decomposition valid for every unit vector v . Use local unitaries to write $v = \sum_{i=1}^r s_i |ii\rangle$, with $s_i \geq 0$ and $\sum_i s_i^2 = 1$. Let the independent phases θ_i be uniform on $\{0, 2\pi/3, 4\pi/3\}$ and set

$$x_\theta = \sum_i \sqrt{s_i} e^{i\theta_i} |i\rangle, \quad z_\theta = \sum_i \sqrt{s_i} e^{-i\theta_i} |i\rangle.$$

A finite phase average of product projectors gives

$$\Omega = \mathbb{E}_\theta [|x_\theta \otimes z_\theta\rangle\langle x_\theta \otimes z_\theta|] = vv^* + \sum_{i \neq j} s_i s_j |ij\rangle\langle ij|.$$

Indeed, the only nonzero averaged phase factors correspond to identical ordered pairs or to two diagonal pairs. Consequently

$$I + 2vv^* = 2\Omega + \sum_i |ii\rangle\langle ii| + \sum_{i \neq j} (1 - 2s_i s_j) |ij\rangle\langle ij| + I_{\text{out}},$$

where the displayed sums range over the Schmidt-support product subspace and I_{out} is the identity on its complementary product-basis coordinates. Every term is separable and every coefficient is nonnegative because $2s_i s_j \leq s_i^2 + s_j^2 \leq 1$. Thus $(I + 2vv^*)/(3n+2)$ is separable for every unit v . Any global-unitary conjugate has the same form with a new unit vector, proving absolute separability and the required attainment. $\square$

Remark 9.2 (Formalization boundary). Corollary 9.1 is an analytic consequence using the explicitly cited separability criterion and the decomposition above. It is *not* included in the immutable Lean proof of the APPT theorem. No claim is made that the separability cone, the cited criterion, or this corollary has been formalized in that release. In particular, equal maximal purities do not imply equal convex sets.

10 Scope and further questions

The main theorem closes the maximal-purity problem for $3 \otimes n$ with $n \geq 3$. The $3 \otimes 2$ system is, after swapping factors, a qubit–qutrit problem and is not covered by the present formal theorem. Neither the proof nor the release claims all-maximizer classification or the general- m purity conjecture. Corollary 9.1 gives the analogous absolute-separability maximum analytically; the immutable Lean release covers only the APPT theorem.

The arbitrary-dimensional inscribed-polytope purity conjecture of [3] is false by Theorem 6.1. Theorems 7.1 and 7.3 give the corrected first excess-purity term whenever the total dimension grows, while the exact finite-dimensional maximum and its equality cases remain open. A separate, more structural question is whether absolute PPT and absolute separability coincide beyond qubit–qutrit systems. The additional witness analysis in Section 9 establishes a common qutrit maximum, but proves no reverse inclusion between the two sets. Equal purity maxima must not be mistaken for equality of the convex sets.

Author, funding and AI assistance. This research was conducted independently and received no external funding. The institutional affiliation identifies the author’s place of study and does not assert institutional sponsorship or endorsement. AI assistance was used in mathematical exploration, certificate and program development, Lean proof development, verification orchestration and manuscript drafting. Formal acceptance refers to the stated Lean declarations and audited dependency closure; it is not a claim of external peer review or independent human certification of every narrative statement. No separate claim of historical priority is made beyond the dated public artifacts and the explicit comparison with prior work above.

Data and code availability. The exact qutrit data and independent checker accompany this preprint. The complete qutrit proof, reproduction scripts and verification evidence are frozen in [6]; the earlier immutable interim release is left unchanged. The higher-dimensional analytic proof and its standard-library exact checkers are included in the supplementary directory, with the source files and hashes recorded in the repository. This manuscript is a preprint, not a journal-accepted article.

A Certificate file identifiers

The following SHA-256 digests identify the exact certificate data. The ancillary manifest also records the checker and this preprint’s other source files.

Filename and SHA-256
qutrit_cubic_certificate_search.json 0e5fb10986416ab1a8a5420d92a7a689b5a55d063c9a417d6ce53c2960f1ad77
qutrit_D12_cubic_certificate.json 8ebdf6d6b3712018f91818c6d2bf6336d6a7c6befb34245bf12d719fd3dd31cb
qutrit_D15_fast_certificate.json c5f688e354e3d2aeabbde72ea343316df64867d60dd095c6f8ee201b466d3df8
qutrit_D18_fast_certificate.json 1312bfc0488d8f7ca8afff43007d93d62d81dc9a0cd664f57196b4622d328811
qutrit_D21_fast_certificate.json 34931c9638725825a0efe8bd80d4a1cb15e53de6d1b433b8b77809f09597ea46
qutrit_D24_fast_certificate.json b19f93c12f369530ff436f095700efa631b029a8964a29e32fb3ea125d22d65e
qutrit_uniform_parameter_search.json d381b47b3bcb67ddc457f7c866b5a6e97757e81cb9d7a84fb832a7b397a826a3

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- [6] Y. Zhang, Qutrit–qudit APPT purity: complete all- n Lean proof v1, immutable software and proof-data release (2026), `appt-qutrit-purity-complete-v1`. GitHub immutable proof release; publication commit `2fc3f25179cf1128c4bccbde65fd3a07c093ad6d`; successful audit 37968137659.